DEEP LEARNING-DRIVEN IMAGING MASS CYTOMETRY DENOISING: PERFORMANCE EVALUATION AND IMPACTS IN DOWNSTREAM ANALYSIS OF PDAC AND IPMN DATASETS

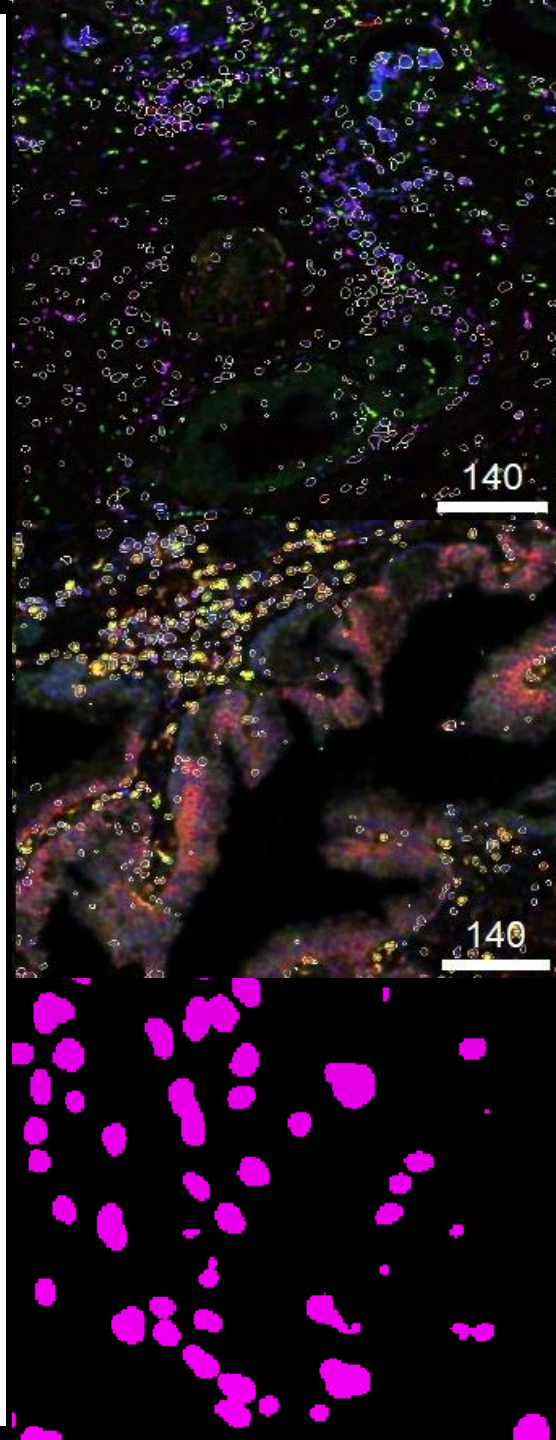
A Master's Thesis by Hugo Gabriel Rodrigues Roca (Maiques Lab)

Submitted to the Barts Cancer Institute (BCI) – The Barts and the London School of Medicine and Dentistry

Queen Mary University of London

MSc Cancer Genomics and Data Science

Barts Cancer Institute
Queen Mary University of London



PANCREATIC CANCER: A SILENT KILLER ON THE RISE

Pancreatic cancer remains one of the deadliest malignancies worldwide

- > 55 years old
- Higher frequency in males
- Incidence is exponentially increasing

Characterised by:

- Late onset + Rapid progression
- Low survival rates + Limited therapeutic responsiveness
- Initially asymptomatic (easily misdiagnosed)
- Commonly arising from pre-cancerous lesions

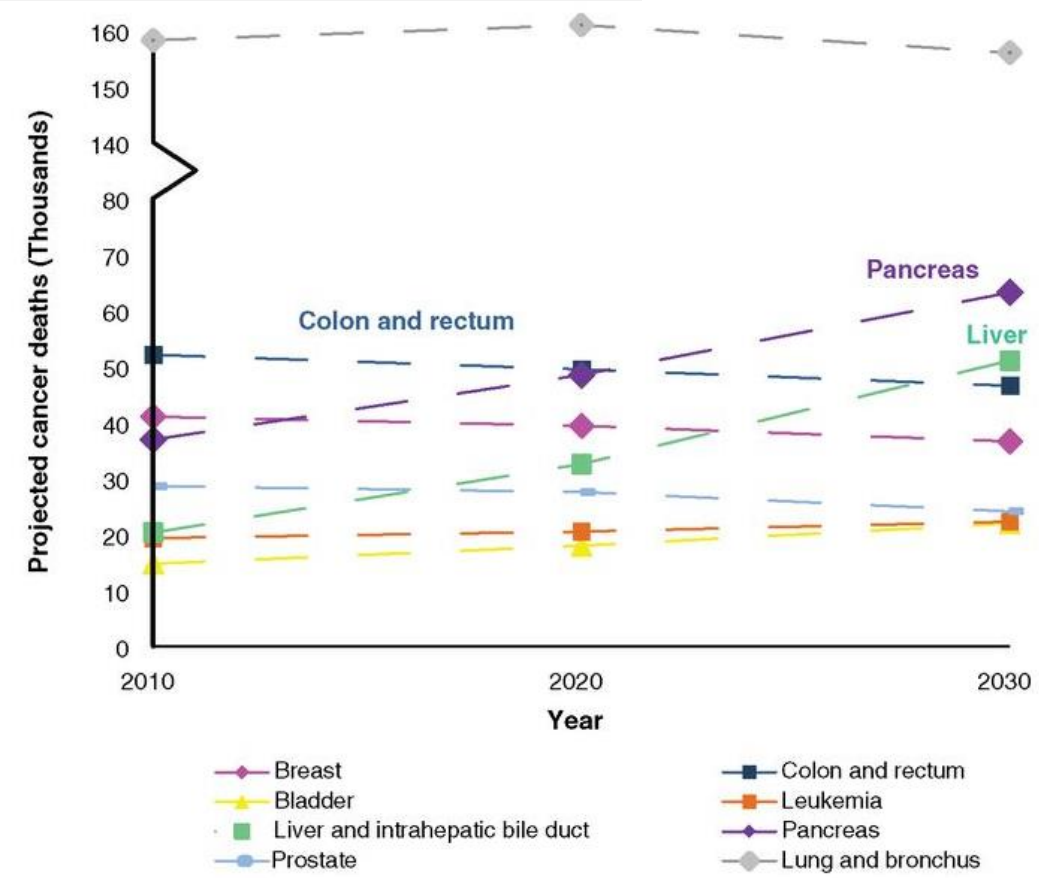


Figure 1) Death projections of the top cancer killers due to demographic changes and the AAPC in death rates (United States).

Adapted from (Rahib et al., 2014).

FROM PRECURSOR LESIONS TO TUMOUR: PANCREATIC CANCER FORMATION

- Pancreatic Intraepithelial Neoplasia (PanIN)
- Intraductal Papillary Mucinous Neoplasm (IPMN)

↑ Risk of Malignancy

PDAC most common type of pancreatic cancer (90% of patients)

➢ **Immune Cold Tumour**

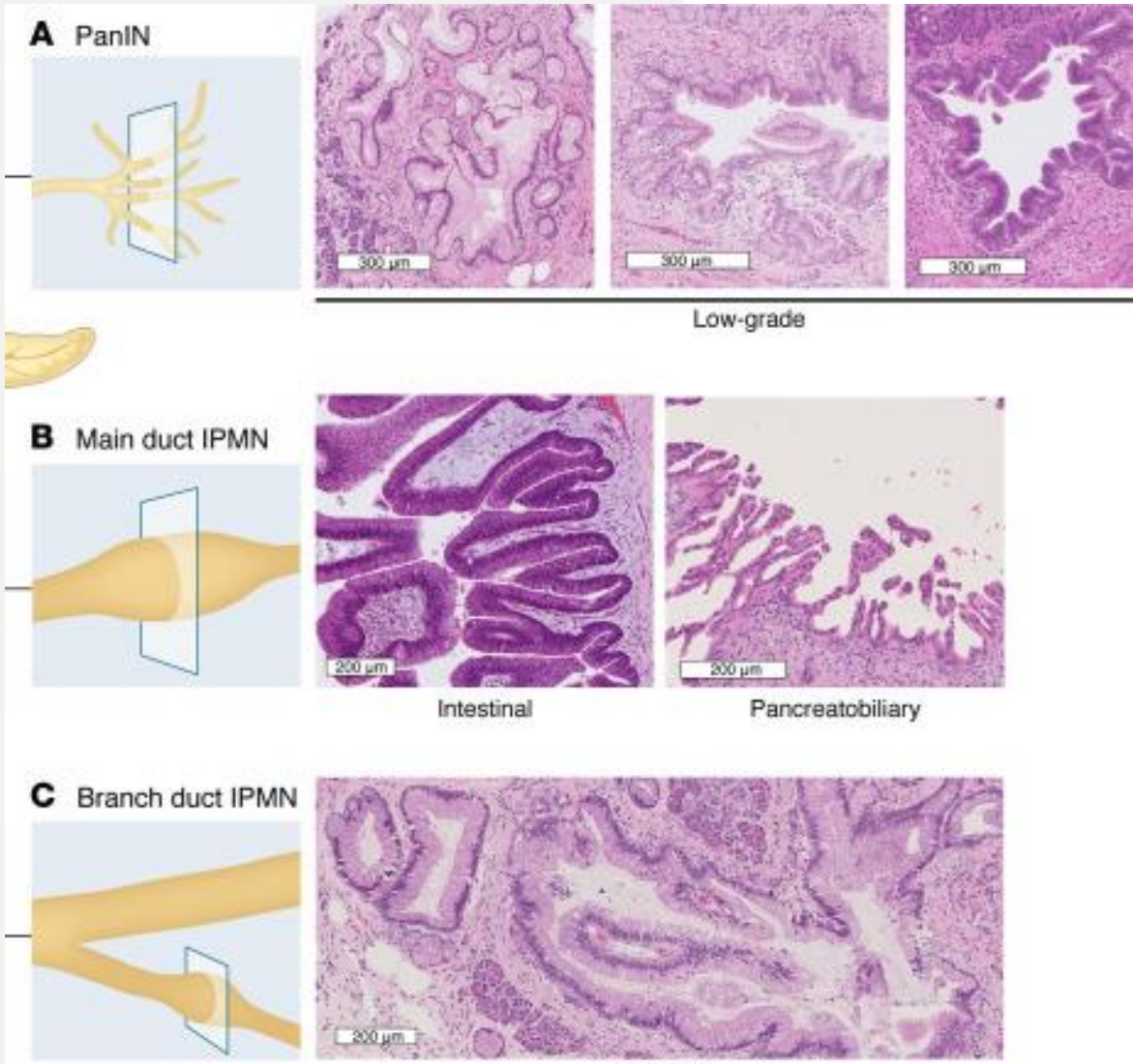


Figure 2) PanIN and IPMN anatomy and histological features.

TUMOUR MICROENVIRONMENT (TME) OF PDAC

- Immune Cold
- Thick Extracellular Matrix (ECM)
  - Highly Fibrotic (CAFs)
  - Collagen-I
- Low immune cell infiltration
  - CD8+ / CD4+ depletion mediated by FoxP3+ Treg cells
- Metabolic Reprogramming
- Highly Metastatic

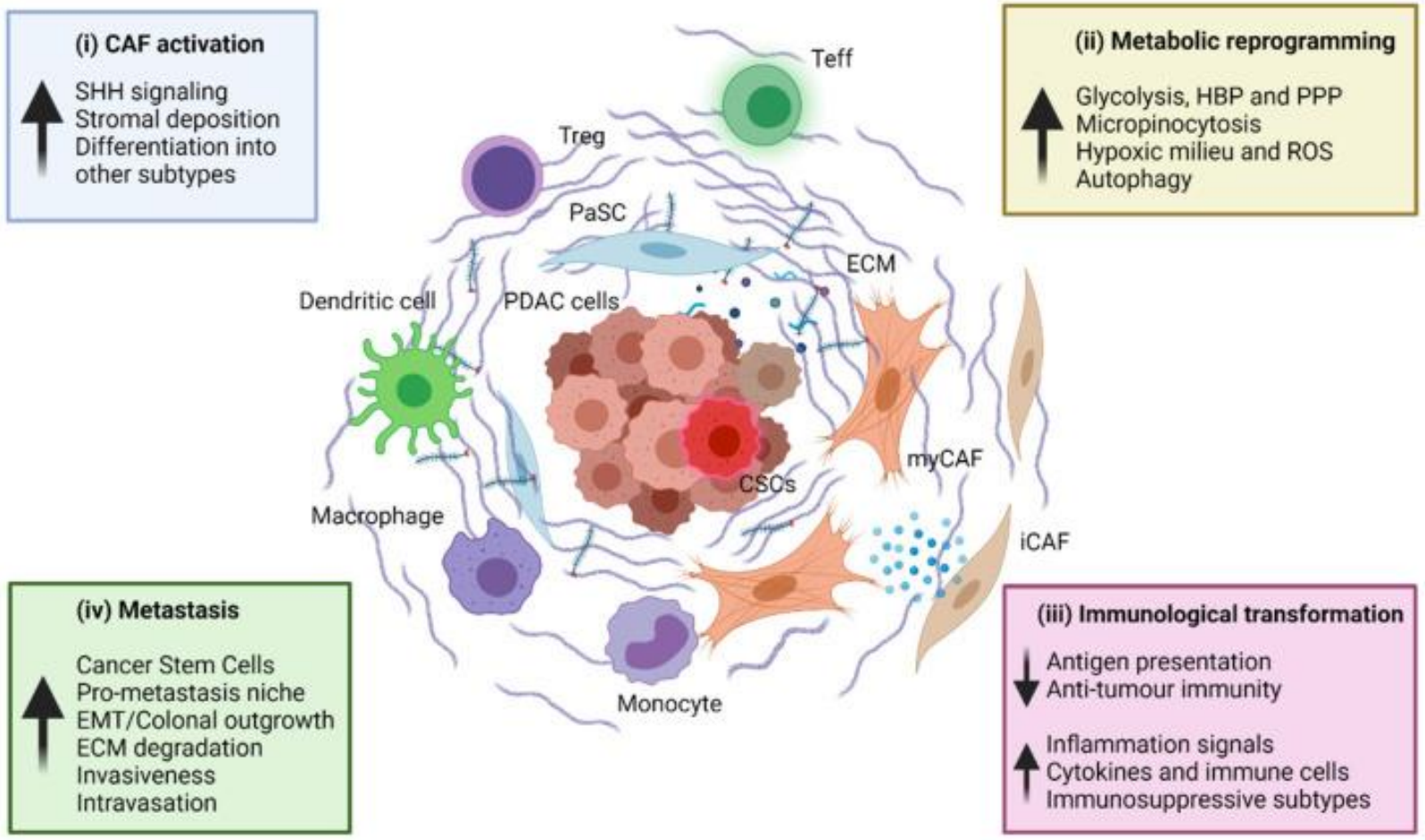
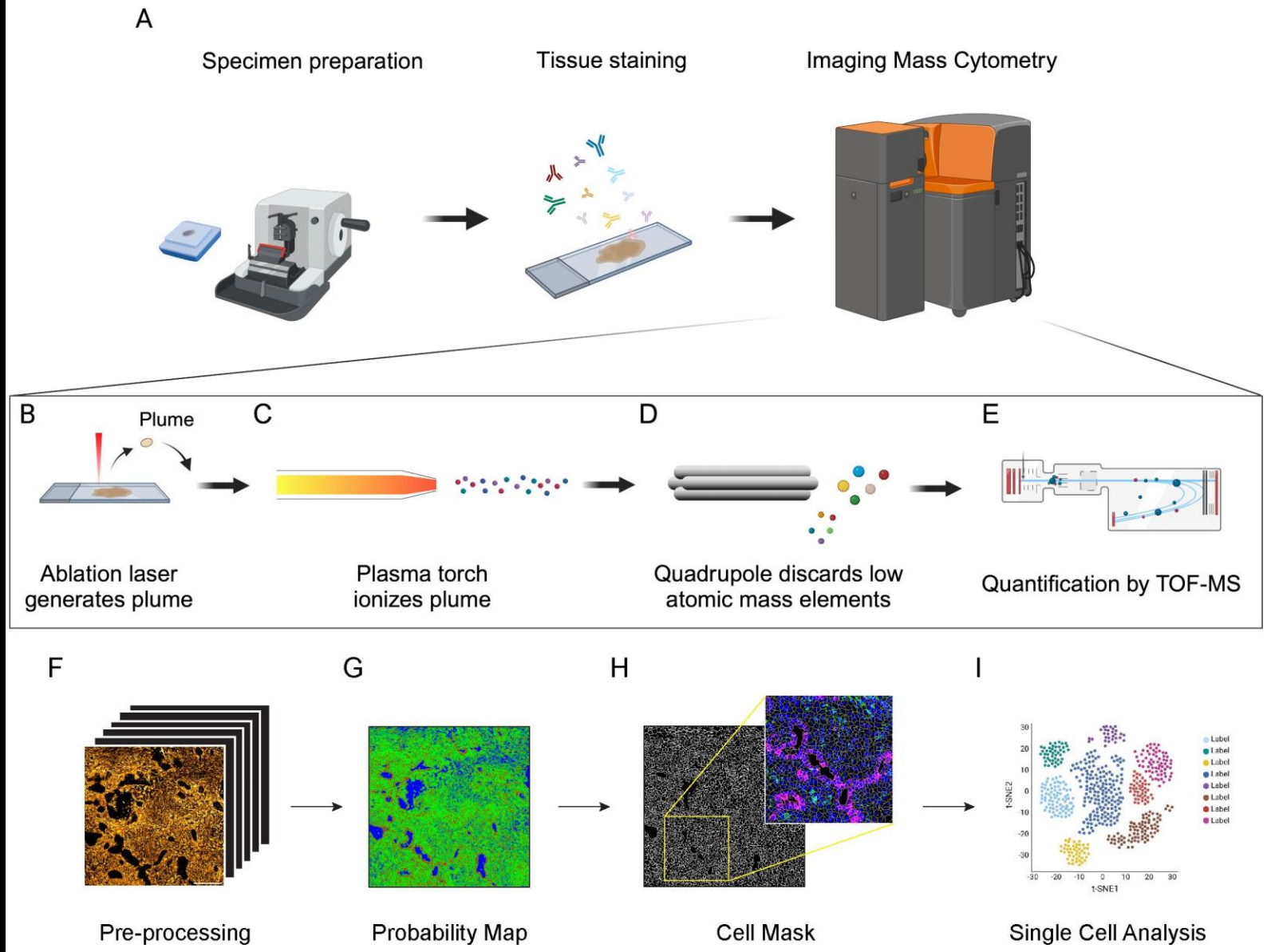


Figure 3) PDAC microenvironment and the cellular interplay. Image extracted from (Truong & Pauklin, 2021).

IMAGING MASS CYTOMETRY: BRIDGING SINGLE-CELL RESOLUTION AND SPATIAL CONTEXT



Challenging factors inherent to this technique:

- Background noise
- Channel spillover
- Hot pixels
- Shot noise
- Staining artefacts
- Low quality samples

Figure 4) Schematic principle of Imaging Mass Cytometry (IMC) technology and downstream analysis. Image extracted from (Erreni et al., 2025).

IMC-DENOISE, A DEEP LEARNING TOOL TO IMPROVE IMC IMAGE QUALITY

<u>IMC-Denoise</u> outperformed existing adaptive hot pixel and background noise removal methods across distinct tissue sections (Figure 5) (Lu et al., 2023).

- Differential Intensity Map-based Restoration (DIMR) - remove hot pixels
- DeepSNiF (DL) - cleans shot noise and background residuals

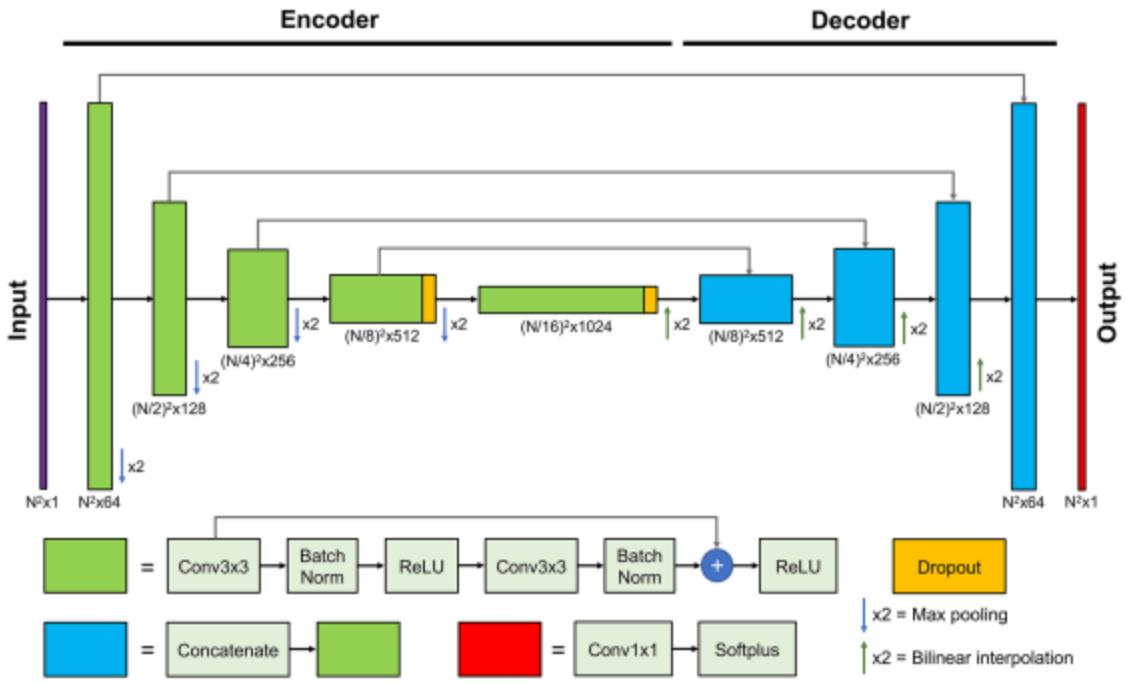


Figure 5) The proposed convolutional neural network structure used in IMC-Denoise. The overall structure of the network follows the U-Net architecture modified with Res-block modules. Image extracted from (Lu et al., 2023).

AIM OF THIS STUDY

Train and evaluate the performance of IMC-Denoise self-supervised DL models

- Effectively reduced noise in PDAC & IPMN IMC samples?
  - Immune / Stromal / Nuclear Markers
- Improves Image Quality?
- Downstream analysis
  - Denoising effects on:
    - Cell Segmentation
    - Expression Marker Quantification

IS THIS APPROACH A VALUABLE OR TRIVIAL METHOD IN IMPROVING IMC IMAGE QUALITY & DOWNSTREAM ANALYSIS FOR STUDYING PANCREATIC CANCER TME?

AIM OF THIS STUDY

Pancreatic histology samples

↓

University of Verona

↓

ROIs assembled into TMA slides

↓

Imaging Mass Cytometry

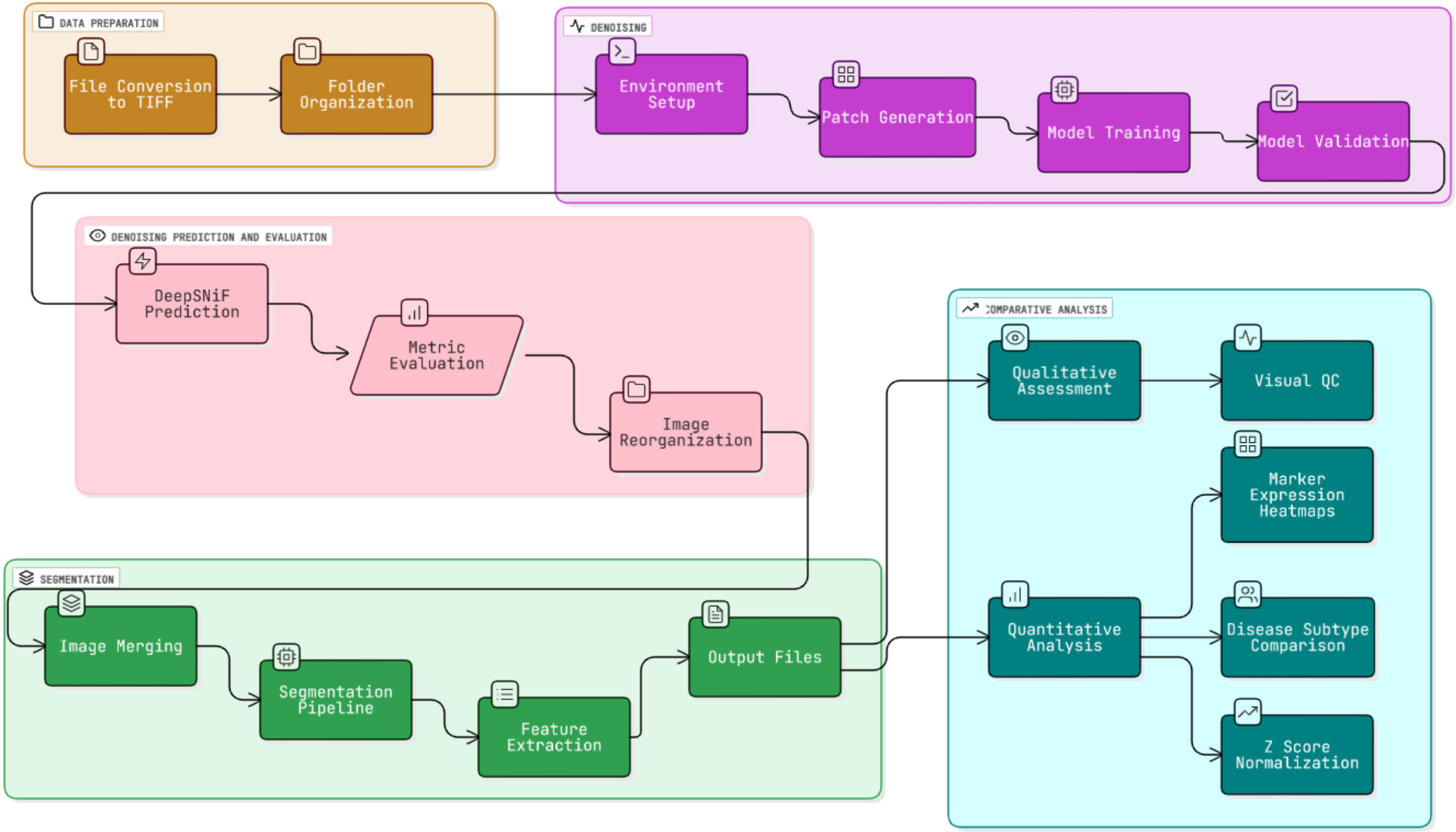
↓

Multi-channelled images (n = 371) PDAC (n = 187) and IPMN (n = 184)

Mass Cytometry Panel

Isotope-tagged markers for IMC

Isotope	Metal	Target Marker
149Sm	Samarium	CD11b
146Nd	Neodymium	CD16
147Sm	Samarium	CD163
161Dy	Dysprosium	CD20
170Er	Erbium	CD3
156Gd	Gadolinium	CD4
152Sm	Samarium	CD45+
166Er	Erbium	CD45RA
159Tb	Terbium	CD68
162Dy	Dysprosium	CD8
169Tm	Thulium	Collagen I
155Gd	Gadolinium	FoxP3
167Er	Erbium	Granzyme
173Yb	Ytterbium	CD45RO
193Ir	Iridium	DNA3

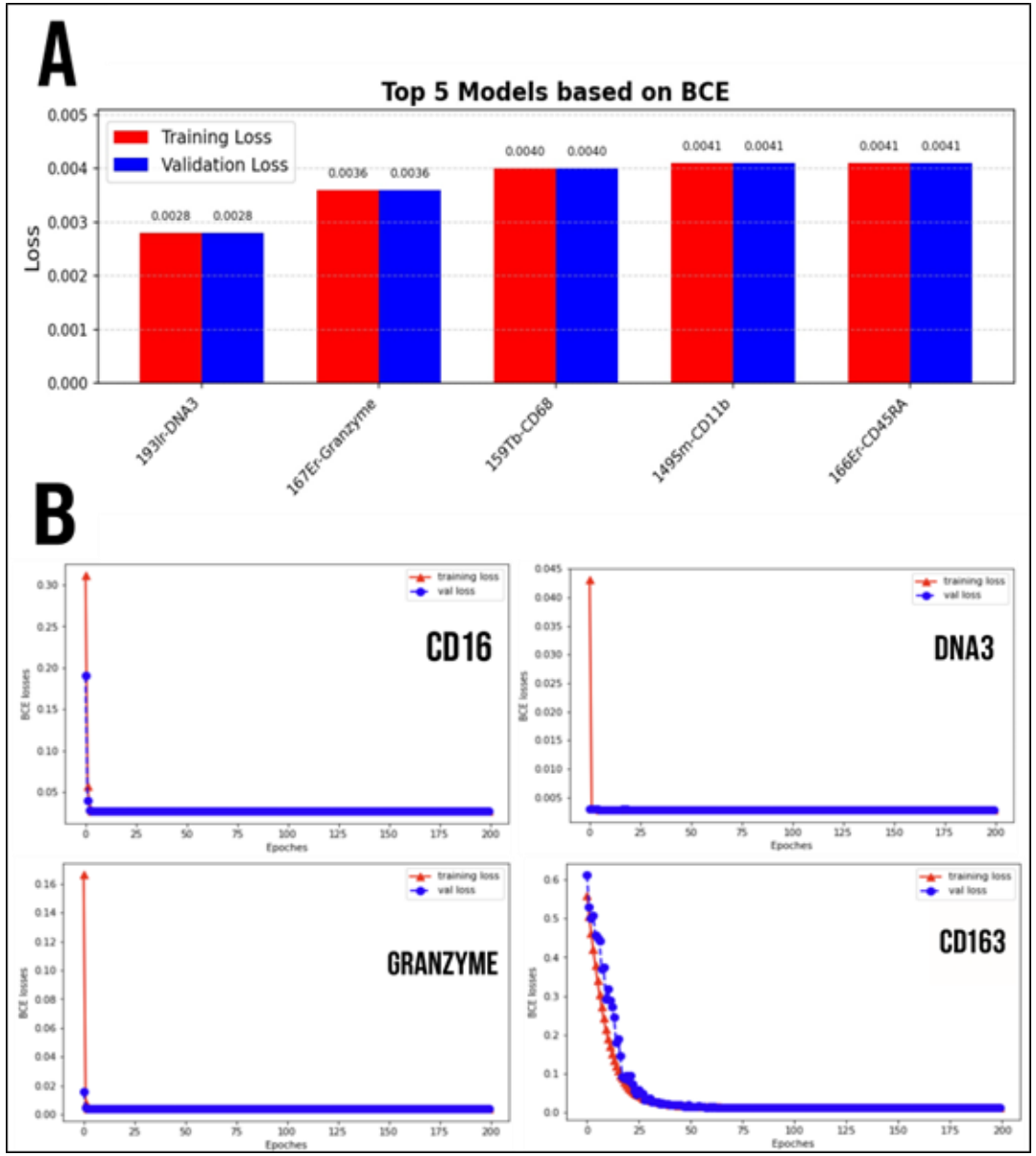

DATA PREPARATION
File Conversion to TIFF
Folder Organization
DENOISING
Environment Setup
Patch Generation
Model Training
Model Validation
DENOISING PREDICTION AND EVALUATION
DeepSNiF Prediction
Metric Evaluation
Image Reorganization
SEGMENTATION
Image Merging
Segmentation Pipeline
Feature Extraction
Output Files
COMPARATIVE ANALYSIS
Qualitative Assessment
Visual QC
Quantitative Analysis
Marker Expression Heatmaps
Disease Subtype Comparison
Z Score Normalization

RESULTS – MODEL TRAINING PERFORMANCE

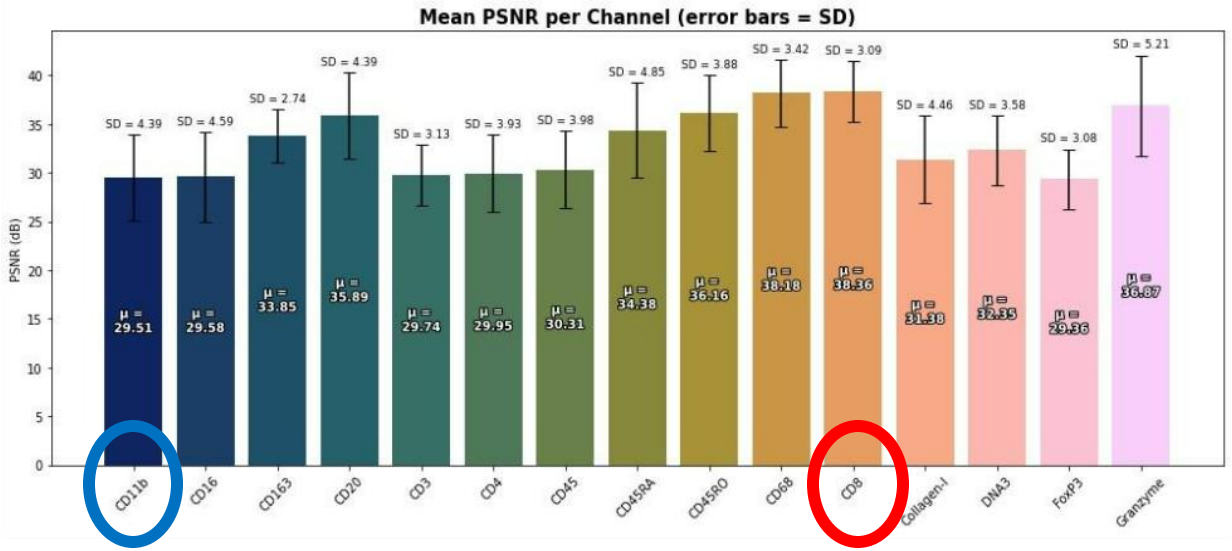
IMC_Denoise Channel Training and Validation Final Losses

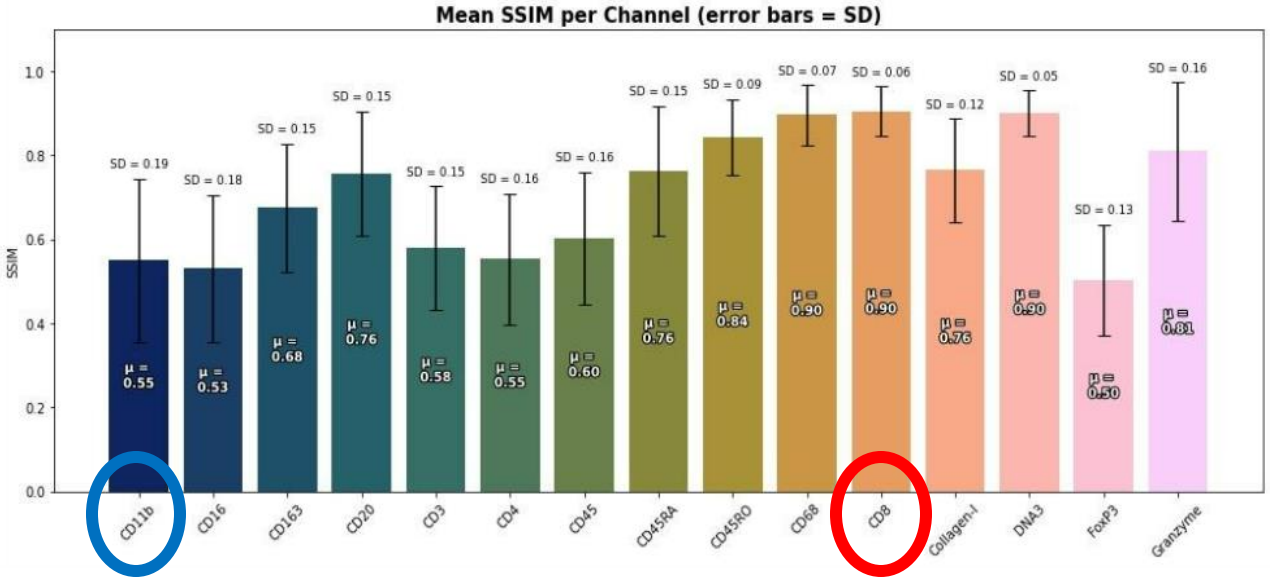
Channel	Training Loss	Validation Loss
149Sm-CD11b	0.0041	0.0041
146Nd-CD16	0.0266	0.0267
147Sm-CD163	0.0119	0.0111
161Dy-CD20	0.0099	0.0101
170Er-CD3	0.0264	0.0256
156Gd-CD4	0.0217	0.0219
152Sm-CD45+	0.0173	0.0172
166Er-CD45RA	0.0041	0.0041
159Tb-CD68	0.004	0.004
162Dy-CD8	0.0093	0.0095
169Tm-Coll	0.0056	0.0056
155Gd-FoxP3	0.0222	0.0222
167Er-Granzyme	0.0036	0.0036
173Yb-CD45RO	0.0051	0.0051
193Ir-DNA3	0.0028	0.0028

Loss values generated from 85% of images used for Training and 15% for Validation (n = 371)

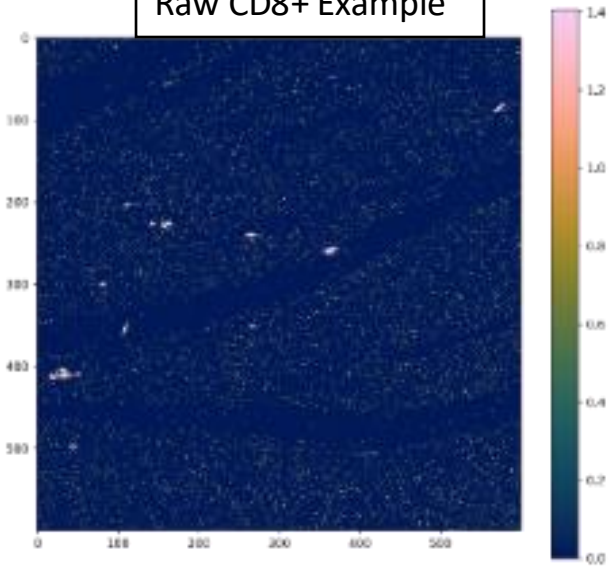


RESULTS – DENOISING EVALUATION

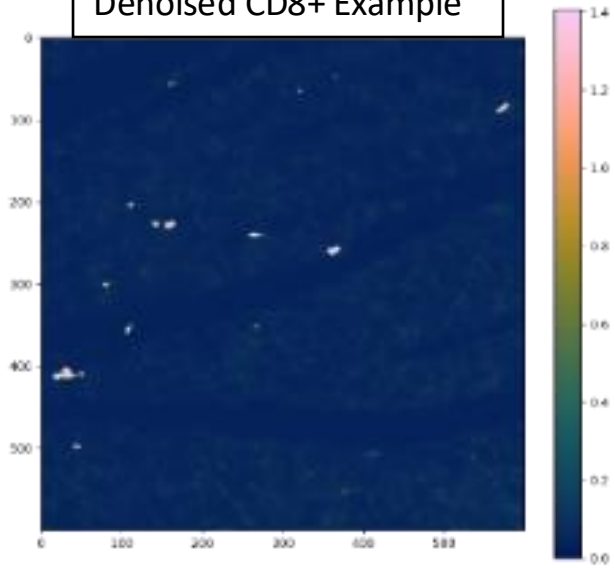




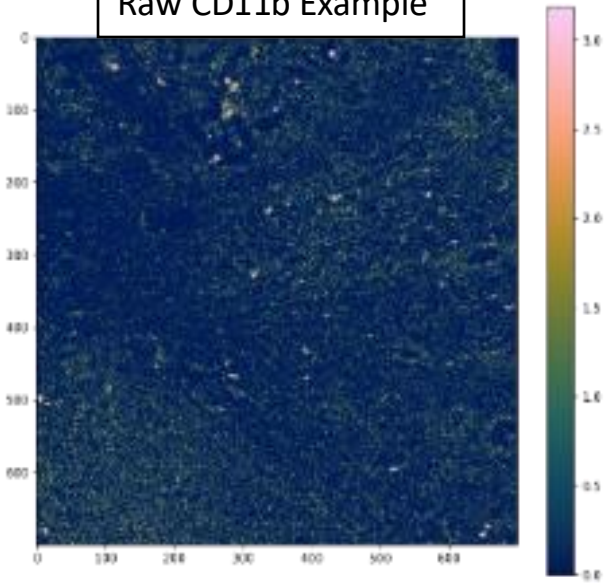
Raw CD8+ Example



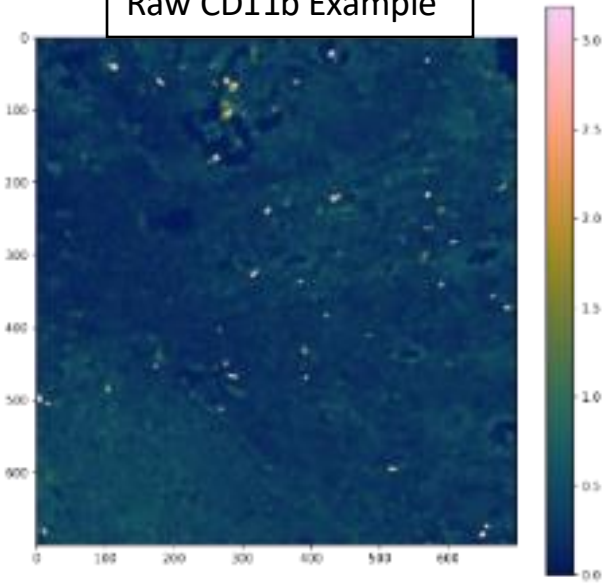
Denoised CD8+ Example



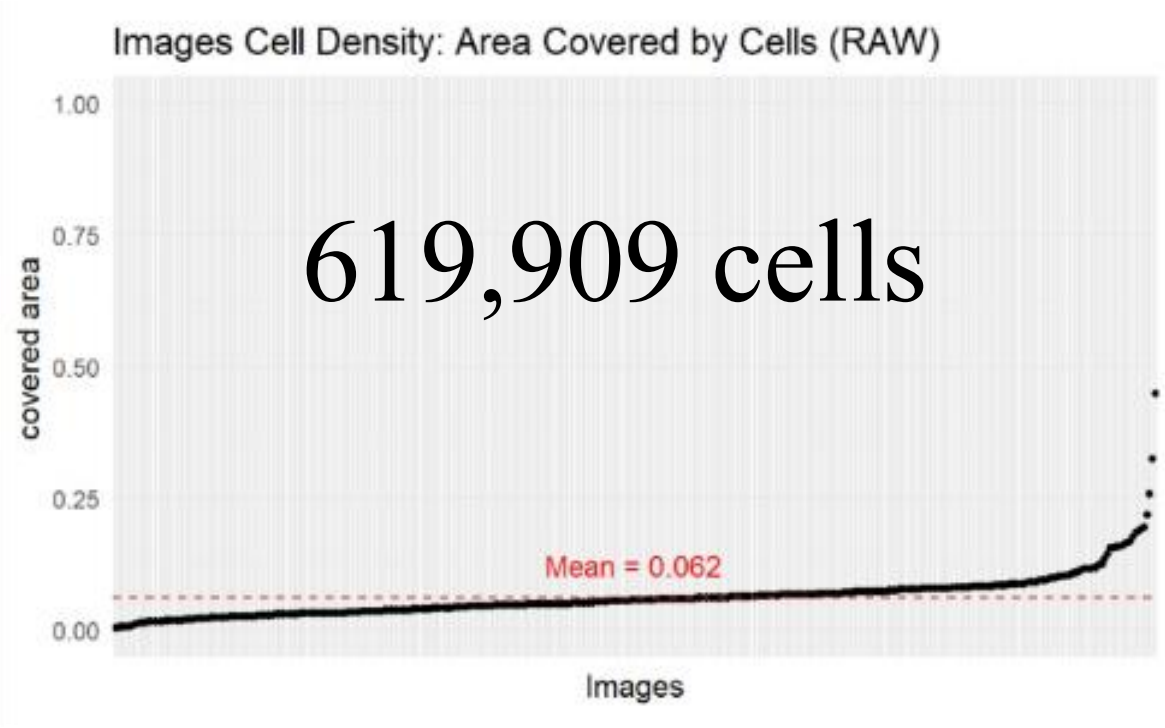
Raw CD11b Example

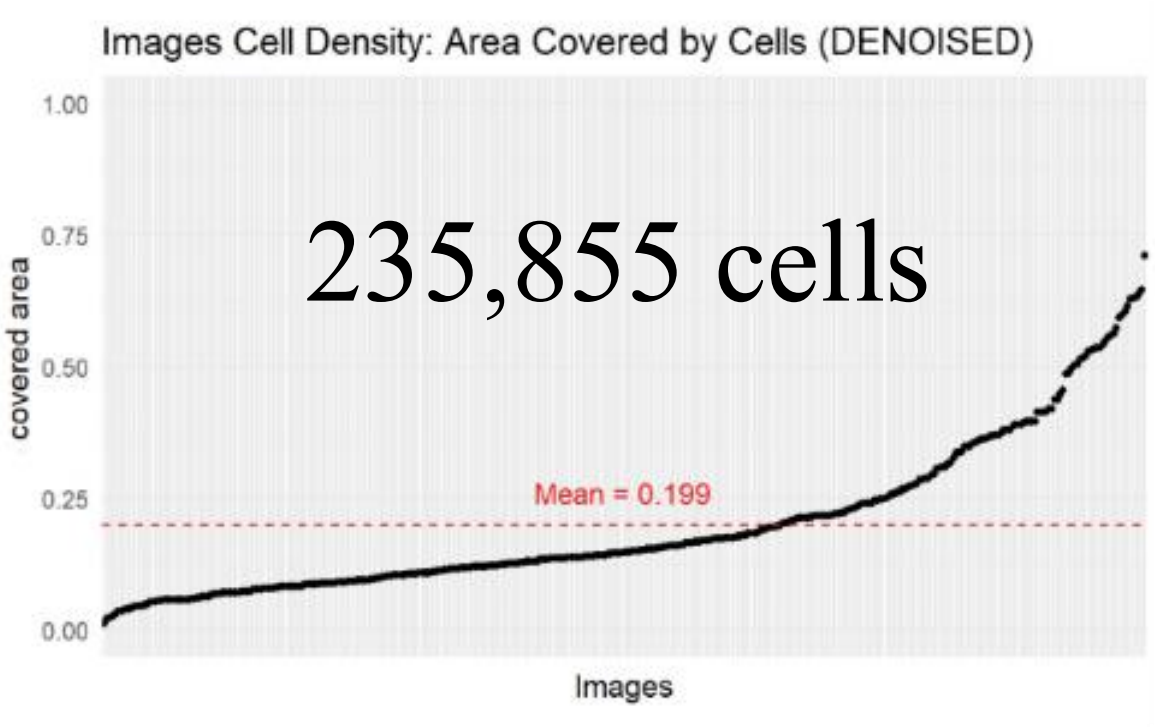


Raw CD11b Example

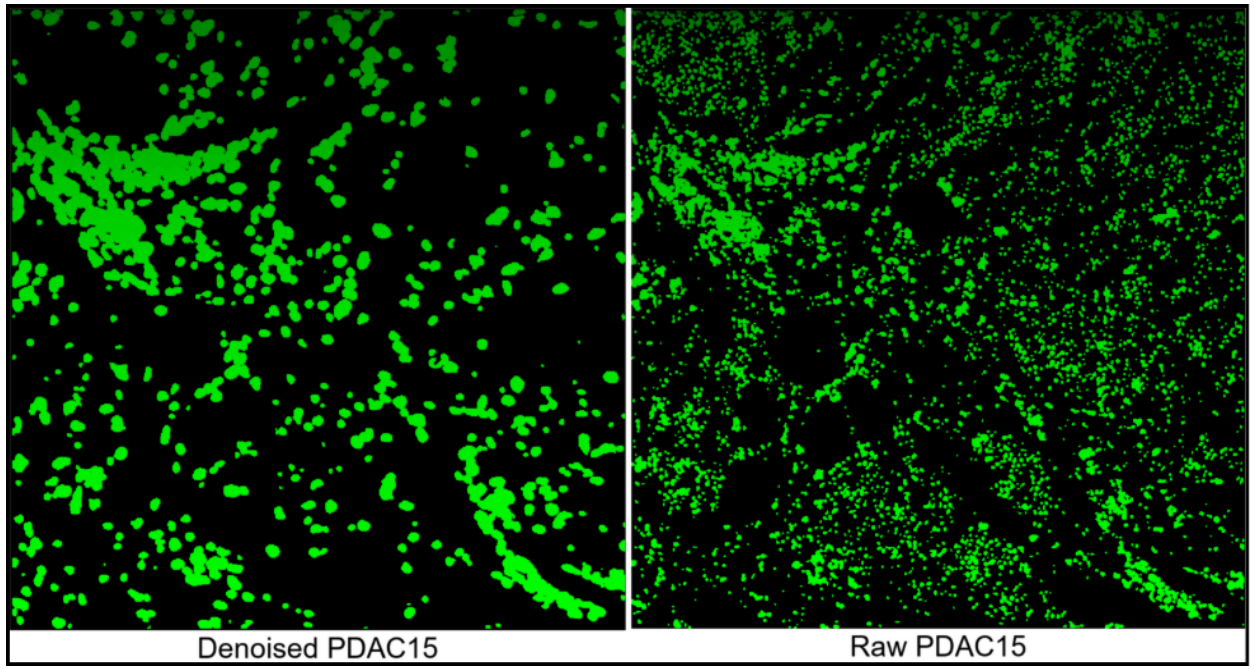


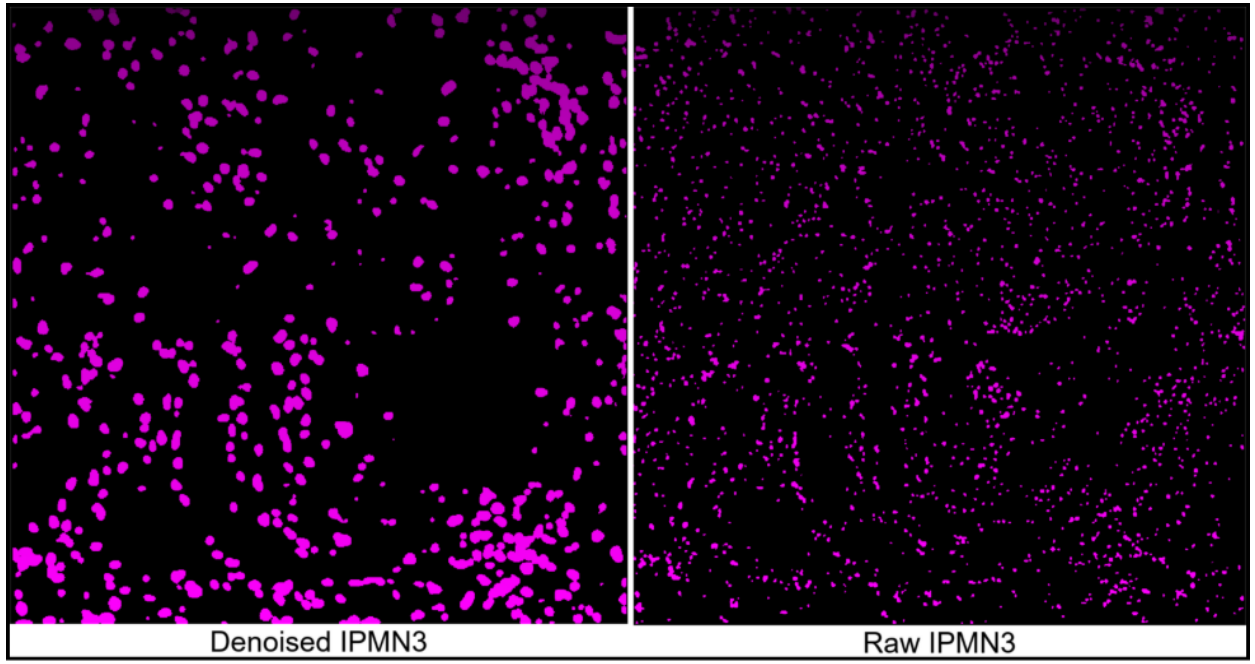
RESULTS – STEINBOCK SEGMENTATION

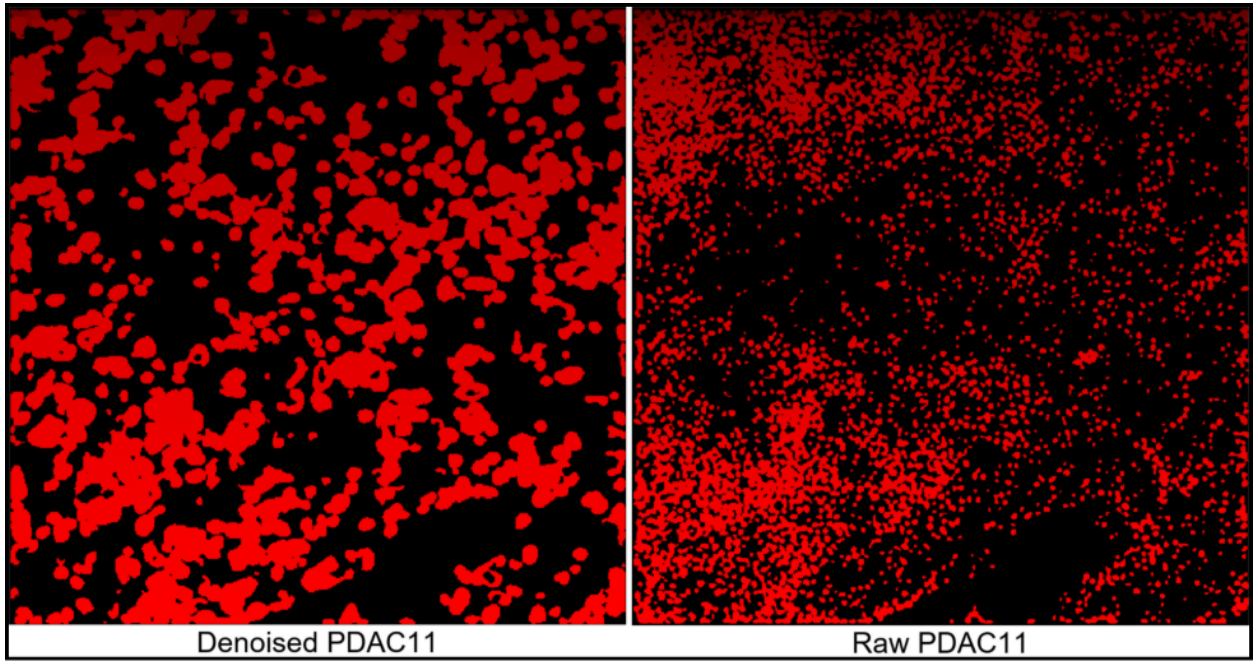




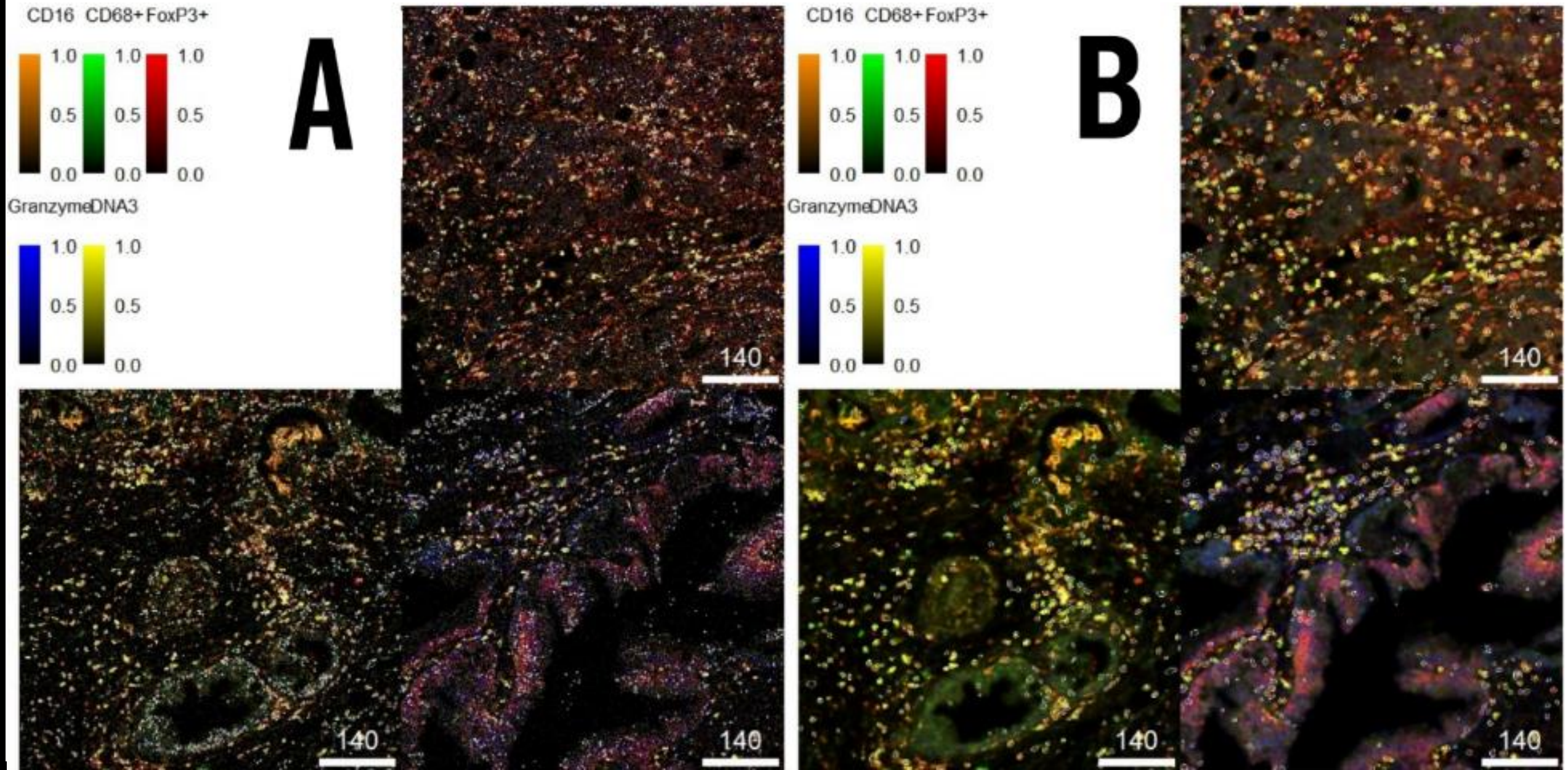
RESULTS – RAW VS DENOISED CELL MASKS



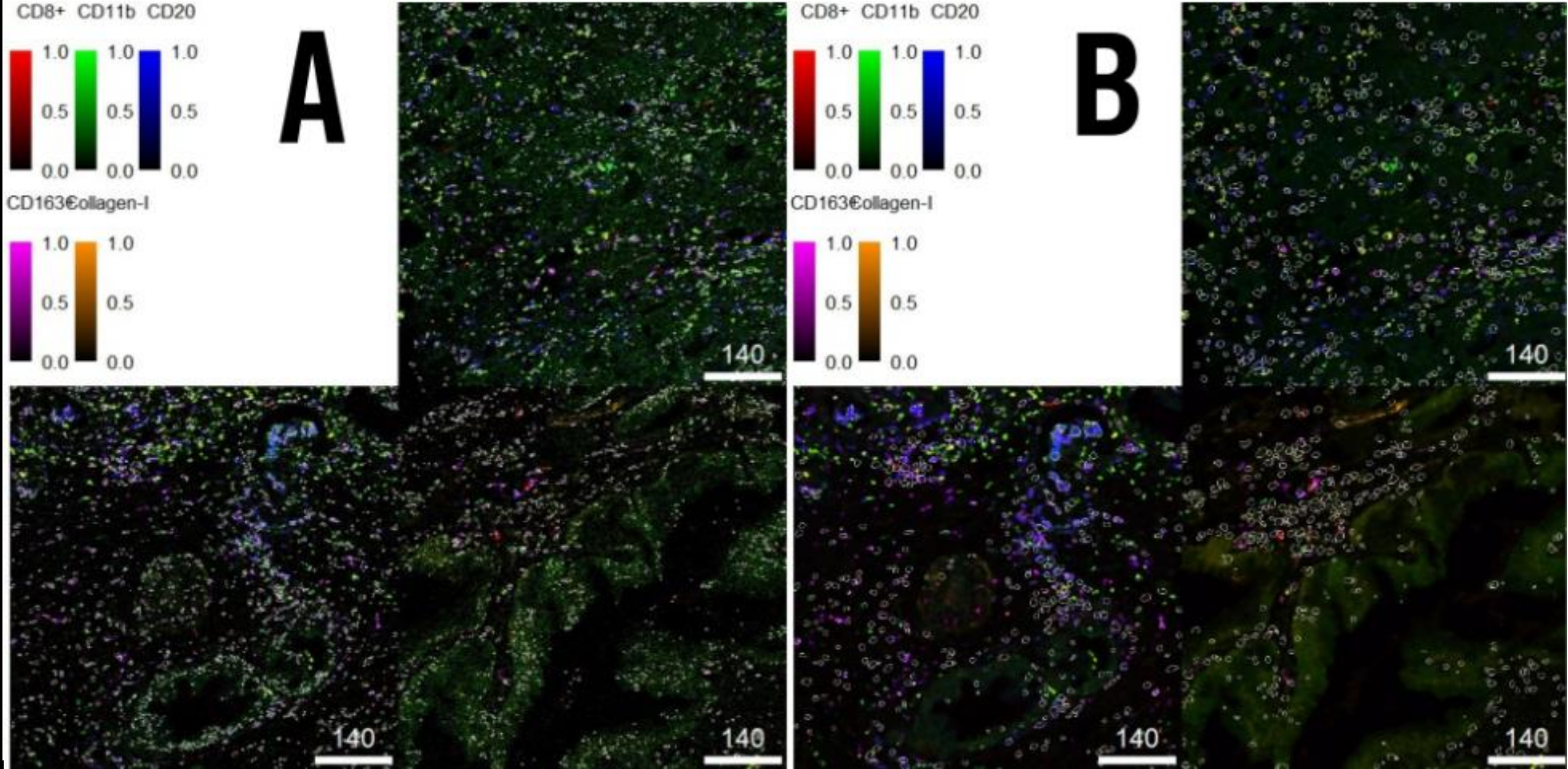




RESULTS – CELL MARKER VISUALISATION

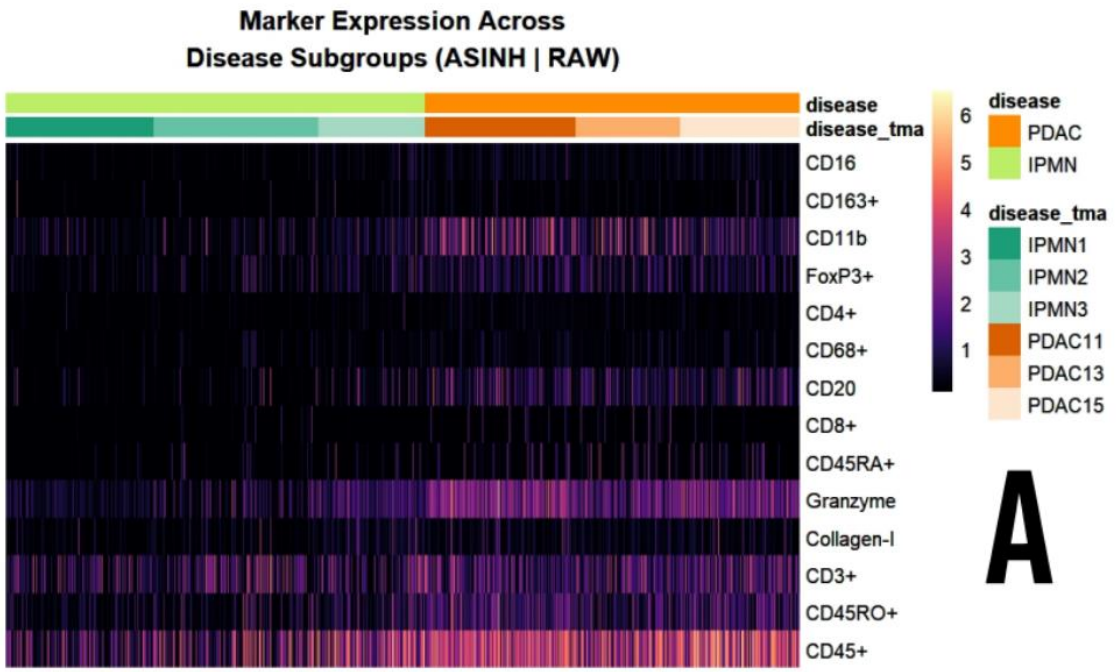


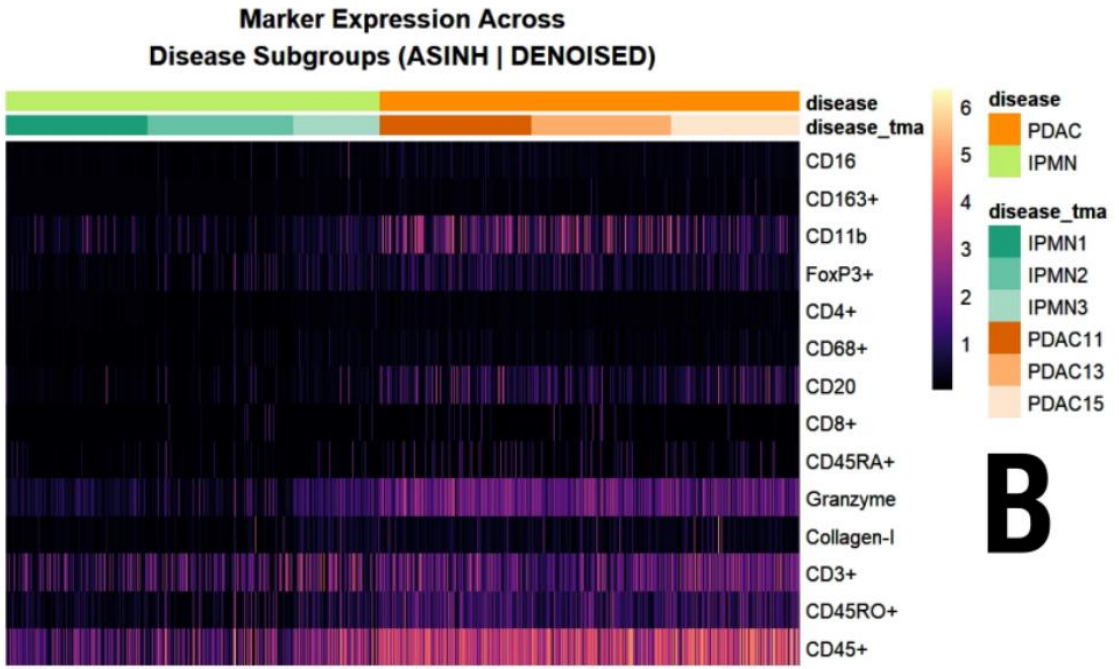
RESULTS – CELL MARKER VISUALISATION

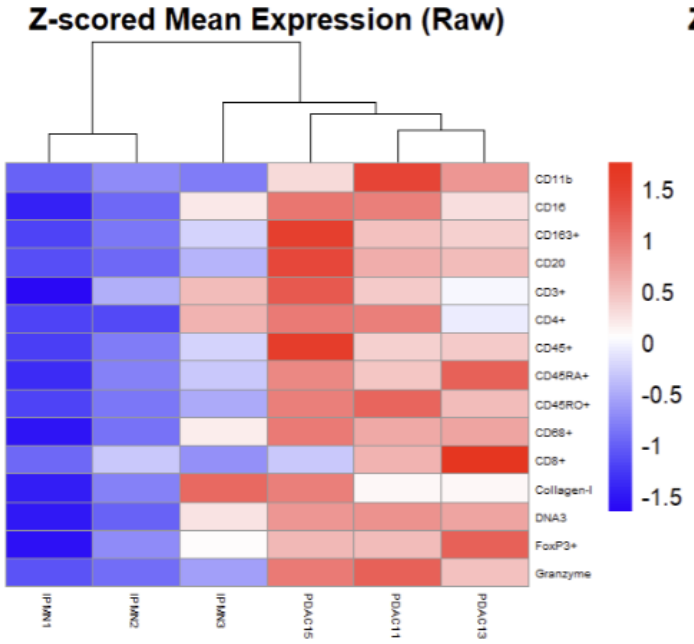


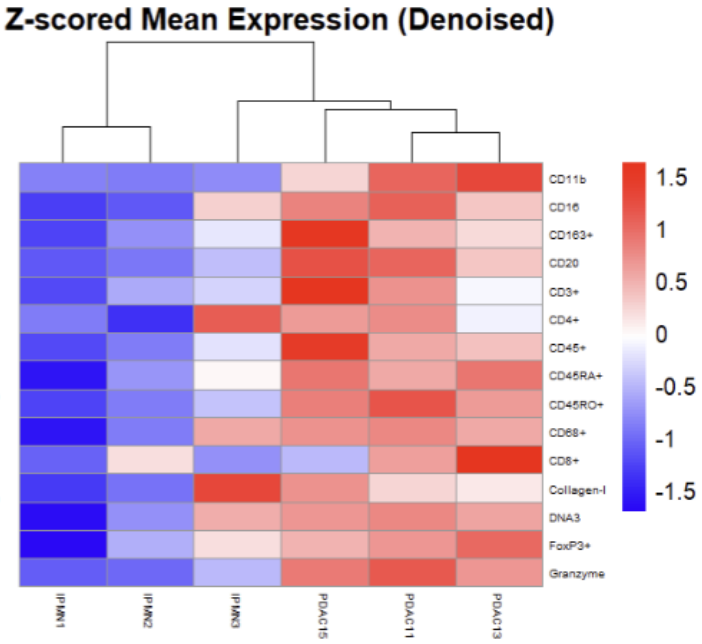
Marker Expression Analysis Post Segmentation

- Tune noisy channels and remove false positives
- Enhance relation between FoxP3^{+} -> CD8^{+} -> CD11b^{+}axis
- Enhance CD20+ B-cell cluster recognition
- Improve image quality and marker visualisation
- Without shifting significantly biology of expression patterns









THANK YOU

Any questions?